

A Three-Layer Analysis of Commodity Price Distributions
Under Geopolitical Stress
Evidence from Middle East Escalation Scenarios

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Iran War Shock: Geopolitical Risk and Commodity Prices

The Strait of Hormuz is the world’s single most critical commodity choke-point: ~21 million bbl/day of crude & products (≈21% of global supply) plus ~10 bcf/day of LNG (≈20% of global LNG trade) pass through a 33 km-wide passage. A partial closure in late February 2026 triggered a cascade across four sectors: **energy** → **fertilizer** → **food**, with metals exposed independently via energy-intensive smelting and freight disruption.

Commodity Prices Surge since Strait of Hormuz Shock		
Sector	March 2026 Change	Threatened Exposure
Energy	Crude +41%, Brent +46%, TTF +59%, LNG +31%	Power generation, transport, petrochemicals
Fertilizers	DAP +5%	Global grain production
Agriculture	Wheat +7%, Maize +2%	Food-importing nations; live-stock feed chains
Metals	Aluminium +10%	Energy-intensive smelting; Hormuz freight routes

Source: World Bank Pink Sheet (updated April 2, 2026). Monthly average price change February → March 2026.

Analysing the Shock: A Three-Layer Decomposition

Three separable, estimable layers:

Layer	Model	What it adds
Mean $\mu_{i,t}$	ARX-Koyck + sparse network	GPR-conditional price path; cross-commodity propagation
Variance $\sigma_{i,t}$	GJR-GARCH	Asymmetric volatility dynamics
Correlation $\bar{R}$	DCC-MIDAS	Shock co-movement

$f(\mathbf{P}_{t+h} \mid \text{GPR}_t)$ decomposes into three estimable objects: a conditional mean (ARX-Koyck), a conditional variance (GJR-GARCH), and a conditional correlation (DCC-MIDAS). Each layer is independently identified and separately interpretable.

Data Sources

Commodity Prices — World Bank Pink Sheet

14 series chosen for direct Strait of Hormuz exposure; monthly nominal USD, 1960–2026.

Group	Series
Energy	Crude Oil, Brent, NatGas US, NatGas EU, LNG Japan, Coal AU
Fertilizers	Urea, DAP, TSP, Phosphate Rock
Agriculture	Wheat, Maize
Metals	Copper, Aluminum

GPR Index — Caldara–Iacoviello (Caldara and Iacoviello 2022)

Monthly, 1985–2026.

Trade Balance — UN Comtrade

Bilateral flows 2021–2023 (three-year average).

The GPR Index: Threats vs Acts (Caldara and Iacoviello 2022)

- **GPR**: share of newspaper articles mentioning geopolitical tension, normalized to average 100 over the 1985–2019 period; monthly, 1985–present

GPRT — Geopolitical Threats

Forward-looking escalation risk: war risk, military mobilization, nuclear mentions.

Captures *risk premia* and precautionary demand.

GPRA — Geopolitical Acts

Realized events: military operations, terrorist attacks.

Captures *demand destruction* and recession fears.

Historical anchors

Event	GPRT	GPRA
9/11 — 2001	≈ 180	≈ 410
Iraq War — 2003	≈ 222	≈ 130
Ukraine — Feb 2022	≈ 200	≈ 180
Iran War Mar 2026	293	404

The ARX-Koyck Model

For each commodity i , monthly log-returns:

$$\begin{aligned}\Delta \log P_{i,t} = & \alpha_i + \phi_i \Delta \log P_{i,t-1} + \kappa_i^T \Delta \text{GPRT}_t^* + \delta_i^T \text{GPRT}_{t-1}^* \\ & + \kappa_i^A \Delta \text{GPRA}_t^* + \delta_i^A \text{GPRA}_{t-1}^* + \sum_j \beta_{ij} \Delta \log P_{j,t-1} + \varepsilon_{i,t}\end{aligned}$$

Koyck reparameterization — two distinct GPR effects:

- κ_i^T : coefficient on ΔGPRT_t^* — immediate price response to a *new* GPR surprise this month
- δ_i^T : coefficient on GPRT_{t-1}^* — effect of the *sustained level* of geopolitical tension

Empirical finding: $\delta_i \approx 0$ for most commodities — prices respond to GPR *news* (ΔGPRT), not a permanently elevated index level. Exception: crude oil and Brent show a small but significant negative level effect ($\hat{\delta}^T = -0.025^{***}$), consistent with market normalisation under prolonged tension.

The κ terms are the scenario engine

They differentiate De-escalation, Baseline, and Escalation because each scenario generates a different GPRT path and therefore a different monthly sequence of ΔGPRT . Without the Koyck impulse, all three scenarios produce near-identical mean paths.

Sparse Cross-Commodity Price Network

- Each commodity may load on lagged returns of other commodities — the cross-beta matrix $\{\beta_{ij}\}$ is selected by BIC stepwise
- BIC penalizes additional parameters, retaining only cross-lags with genuine predictive content

Key selected links (all $p < 0.01$):

Link	$\hat{\beta}_{ij}$
TSP $\leftarrow$ DAP	+0.299
DAP $\leftarrow$ TSP	+0.262
TSP $\leftarrow$ Copper	+0.169
DAP $\leftarrow$ Aluminum	+0.168
Phosphate $\leftarrow$ TSP	+0.274

Fertilizer indirect channel

Fertilizer κ^T is not significant — their GPR sensitivity runs entirely through the energy-metals-fertilizer network. The DAP–TSP mutual feedback ($\lambda_1 = 0.76$, stable) is the primary amplification channel.

Estimation Results: GPR Coefficients

Energy commodities:

	$\hat{\phi}$	$\hat{\kappa}^T$	$\hat{\delta}^T$	$\hat{\kappa}^A$	$\hat{\delta}^A$	$\hat{\sigma}$
CrudeOil	0.288***	0.028***	-0.025***	-0.024***	0.016**	0.087
Brent	0.253***	0.028***	-0.025***	-0.024***	0.015**	0.089
NatGas US	0.016	0.010	-0.007	-0.028**	0.024**	0.146
NatGas EU	0.271***	0.011	-0.008	0.002	-0.007	0.092
LNG Japan	0.341***	-0.005	0.005	0.004	-0.005	0.047
Coal AU	0.271***	0.008	-0.003	-0.004	-0.001	0.068

Buy the rumor, sell the news

For crude oil and Brent: $\hat{\kappa}^T > 0$ (threats raise prices via risk premium) and $\hat{\kappa}^A < 0$ (realized acts lower them via demand destruction). GPRA = 404 in March 2026 already signals the “sell the news” dynamic: the conflict has materialized, shifting the dominant channel from risk premium to demand-destruction.

Estimation Results: GPR Coefficients (Other Commodities)

	$\hat{\phi}$	$\hat{\kappa}^T$	$\hat{\delta}^T$	$\hat{\kappa}^A$	$\hat{\delta}^A$	R^2
Urea	0.1466***	-0.0068	0.0091	0.0072	-0.0080	0.024
DAP	0.4331***	0.0046	-0.0007	0.0001	-0.0010	0.192
TSP	0.5222***	0.0062	-0.0015	-0.0039	0.0016	0.278
Phosphate	0.2228***	-0.0045	0.0064	0.0010	-0.0044	0.052
Wheat	0.2331***	0.0058	-0.0044	-0.0035	0.0042	0.056
Maize	0.2722***	0.0086*	-0.0067	-0.0081*	0.0062	0.080
Copper	0.3778***	0.0070	-0.0062	-0.0074	0.0079*	0.149
Aluminum	0.1872***	0.0123***	-0.0076*	-0.0077*	0.0051	0.052

Fertilizers: no direct GPR channel

Urea, DAP, TSP show no significant $\hat{\kappa}^T$ or $\hat{\kappa}^A$ — their price channel runs entirely through energy costs via the network, not geopolitical news. Aluminum responds strongly to threats ($p = 0.003$). Maize shows marginal significance on both GPRT and GPRA.

BIC Model Comparison

Group	Commodity	AR(1)	Koyck-ARX	+Network	Δ BIC	LR p	GPR sig.
Energy	Crude Oil	-970.0	-966.0	-968.7	-2.6	0.0002***	✓
Energy	Brent	-941.3	-933.8	-936.6	-2.9	0.0004***	✓
Energy	NatGas US	-472.3	-463.6	-473.7	-10.1	0.1921	✓
Energy	NatGas EU	-929.8	-903.4	-920.1	-16.7	0.1910	×
Energy	LNG Japan	-1602.8	-1563.6	-1623.6	-60.0	0.6505	×
Energy	Coal AU	-1229.5	-1219.2	-1227.1	-7.9	0.3812	×
Fert.	Urea	-713.8	-690.5	-708.2	-17.7	0.8763	×
Fert.	DAP	-1381.1	-1372.2	-1394.1	-21.9	0.6804	×
Fert.	TSP	-1547.5	-1540.8	-1608.5	-67.7	0.4411	×
Fert.	Phosphate	-1070.8	-1059.5	-1074.4	-14.9	0.8564	×
Agri.	Wheat	-1386.5	-1376.4	-1376.4	0.0	0.7755	×
Agri.	Maize	-1401.7	-1395.2	-1397.3	-2.2	0.3617	✓
Metals	Copper	-1412.6	-1405.4	-1410.3	-4.9	0.4432	✓
Metals	Aluminum	-1468.8	-1464.5	-1472.5	-8.0	0.0642*	✓

LR test: AR(1) vs AR(1)+T+A (joint χ^2_4). GPR sig.: ≥ 1 coefficient $p < 0.10$.

Network adds where direct GPR does not

Δ BIC = +Network minus Koyck-ARX: negative = network preferred. Sparse lags improve BIC for 13/14 commodities (Wheat: zero gain). Note: AR(1) beats both Koyck-ARX and +Network by BIC for 8/14 commodities (including crude oil); the ARX-GPR specification is imposed on theoretical grounds and justified by the significant $\hat{\kappa}$ estimates. Largest network gains: TSP ($\Delta = -68$), LNG Japan (-60), DAP (-22).

Volatility Model: GJR-GARCH

Conditional variance of commodity i :

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \gamma_i \varepsilon_{i,t-1}^2 \mathbf{1}[\varepsilon_{i,t-1} < 0] + \beta_i \sigma_{i,t-1}^2$$

- α_i : ARCH — immediate volatility response to any shock
- γ_i : GJR asymmetry — extra response when $\varepsilon < 0$ (negative surprise raises forward uncertainty more than equivalent positive shock)
- β_i : GARCH persistence — elevated volatility persists for months after a shock

Persistence $\alpha_i + \beta_i + \gamma_i/2$ estimates at 0.85–0.95 across the panel: volatility shocks persist 4–14 months.

Layer 2 of the three-layer model

The GJR-GARCH conditional variance $\sigma_{i,t}$ determines the fan chart width. The current elevated-GPR regime (March 2026) already conditions a wider distribution than the historical average — independently of the scenario path chosen.

GJR-GARCH: Estimated Persistence

Selected GARCH persistence estimates ($\alpha + \beta + \gamma/2$):

Commodity	Persistence	Notes
NatGas EU	≈ 0.95	Highest in panel; 2021–22 crisis spike
Crude Oil	≈ 0.92	Elevated since 2020
DAP / TSP	≈ 0.90	Fertilizer vols elevated since 2021
Aluminum	≈ 0.88	—
Wheat	≈ 0.85	Lower persistence

NatGas EU conditional volatility is used as an additional MIDAS driver in the DCC-MIDAS estimation (alongside GPRT).

Cross-Commodity Shock Correlation: Testing the Independence Assumption

- Standard stress-test models draw commodity shocks **independently** — each market moves on its own
- We test this assumption directly via **DCC-MIDAS** across 14 commodities, conditioning long-run correlation on GPR
- Finding: $\hat{\beta}_T = -0.012$ (LR $p < 0.05$) — **higher geopolitical risk lowers average cross-commodity correlation**
- Mechanism: crises are sector-specific supply shocks — energy spikes, industrial metals decline on recession fears, agriculture responds idiosyncratically to conflict geography
- Implication: the independence assumption is approximately correct for *cross-sector* pairs under stress — geopolitical risk amplifies commodity-specific volatility, not systemic co-movement

Counter-intuitive finding

Geopolitical crises decouple cross-sector commodity co-movement, contradicting the assumption that crises tighten correlations. The diversification benefit of a multi-commodity portfolio is not eroded, and may be enhanced, under escalation.

- **Plain DCC** (R. Engle 2002): dynamic correlations revert to a constant unconditional matrix $\bar{C}$ — one long-run level for the entire sample
- **Problem:** during geopolitical crises the long-run correlation regime should differ from peacetime
- **DCC-MIDAS:** extends $\bar{C}$ to a **slow-moving, time-varying** function of macro drivers at mixed frequency

The long-run correlation between commodities i and j :

$$\bar{\rho}_{ij,t} = \bar{\rho}_{ij}^{\text{base}} + \beta_{ij}^T \sum_{k=1}^K w_k(\theta) \text{GPRT}_{t-k}^*$$

Near-zero ARCH coefficient

$a \approx 0$ is not a model failure — it means short-run correlation dynamics are dominated by slow persistence ($b = 0.947$). In large panels with low average pairwise correlations, individual monthly cross-products are too noisy to update the latent correlation reliably.

The Beta Polynomial MIDAS Weighting

The MIDAS weight function over $K = 24$ monthly lags:

$$w_k(\theta_1, \theta_2) = \frac{(k/K)^{\theta_1-1}(1-k/K)^{\theta_2-1}}{\sum_{l=1}^K (l/K)^{\theta_1-1}(1-l/K)^{\theta_2-1}}$$

- Flexible: depending on θ_1, θ_2 , weights can peak at short, long, or intermediate lags
- Estimated jointly with the DCC parameters by maximum likelihood
- 24 months of GPRT history enter the model — captures medium-term geopolitical regime, not individual monthly noise
- Ensures long-run correlation responds to **sustained** GPR levels rather than transient spikes

DCC-MIDAS: Estimated Parameters

Parameter	Estimate	Interpretation
a (ARCH)	0.0001	Near-zero: short-run updating absent
b (GARCH)	0.9467	Slow mean-reversion; correlations highly persistent
β_T (MIDAS)	-0.0117	Higher GPR $\Rightarrow$ lower avg correlation
LR test p -value	< 0.05	$\beta_T \neq 0$ statistically significant
Avg $\bar{R}$ (full)	0.085	Low average cross-commodity correlation
$\bar{R}$ scenario range	0.083–0.088	Narrow: correlation effect is modest

Why does higher GPR reduce correlation?

Supply disruptions are *sector-specific*: energy spikes, industrial metals decline on recession fears, and agriculture responds to its own geography. The common demand factor weakens as idiosyncratic supply factors dominate.

Implication for tail risk

The DCC-MIDAS $\beta_T = -0.012$ implies geopolitical stress amplifies commodity-specific volatility rather than systemic co-movement. Correlation's contribution to fan-chart width is modest because the ARX cross-betas have already absorbed the predictable co-movement.

Geopolitical Scenarios

Three 12-month GPR trajectories. Starting point: $\text{GPRT}_0 = 293$, $\text{GPRA}_0 = 404$ (March 2026 — Iran War outbreak).

De-escalation Ceasefire by Q3 2026, Strait of Hormuz fully reopens. GPRT decays from 293 towards 103 (historical mean); GPRA normalises rapidly.

Baseline Conflict persists at elevated intensity. GPRT settles at ≈ 200 (Iraq-War-sustained level); GPRA at ≈ 120 .

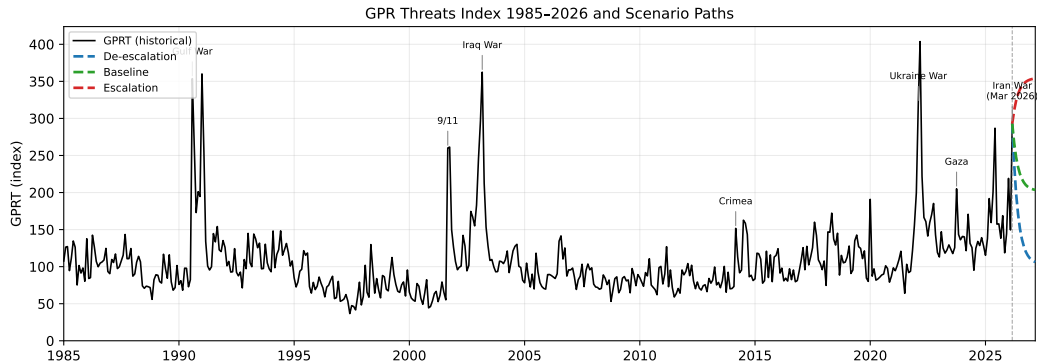
Escalation Regional conflagration: broader military involvement. GPRT rises to ≈ 350 (above all historical precedents); GPRA sustained at ≈ 300 .

Forecast Method

Point forecasts: iterated conditional mean from ARX-Koyck model.

Fan charts: 5,000 Monte Carlo paths with shocks drawn from $\mathcal{N}(0, \Sigma_h)$ where $\Sigma_h = D_h \bar{R}_h D_h$ (D_h from GJR-GARCH, $\bar{R}_h$ from DCC-MIDAS), yielding 90% predictive intervals.

Scenario Calibration to Historical Events



AR(1) mean-reversion at $\rho = 0.710$ distributes the GPR impulse continuously over the 12-month horizon, avoiding an artificial concentration of the ARX scenario effect at $h = 1$.

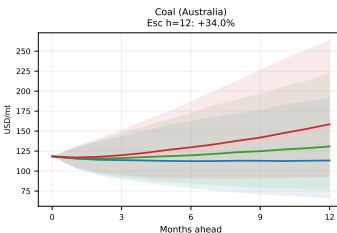
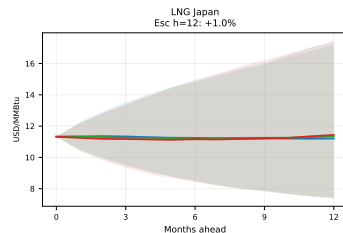
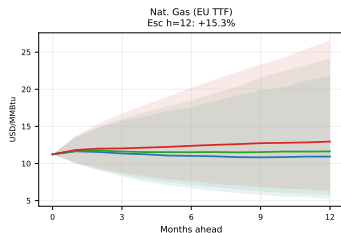
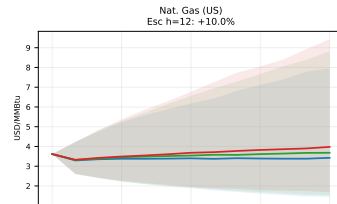
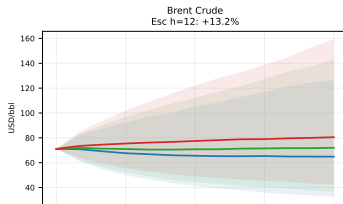
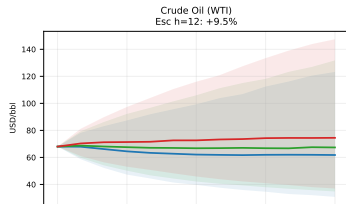
12-Month Price Forecasts: Cross-Scenario Summary

Group	Commodity	Current Mar 2026	De-escal. (%Δ)	Baseline (%Δ)	Escalation (%Δ)
Energy	Crude Oil	95.58	+0.0	+13.0	+32.9
Energy	Brent	103.69	+0.2	+14.9	+38.2
Energy	NatGas US	3.05	+21.3	+30.6	+44.0
Energy	NatGas EU	17.91	+17.8	+35.5	+66.3
Energy	LNG Japan	14.85	+40.6	+49.6	+64.3
Energy	Coal AU	138.60	+15.2	+35.3	+70.7
Fertilizers	Urea	725.63	+30.1	+50.3	+87.2
Fertilizers	DAP	658.25	+54.7	+115.8	+255.7
Fertilizers	TSP	558.13	+59.4	+131.8	+304.6
Fertilizers	Phosphate	152.50	+42.0	+63.0	+99.0
Agriculture	Wheat	275.91	+6.2	+12.2	+22.4
Agriculture	Maize	212.69	+3.4	+10.3	+21.1
Metals	Copper	12,529	+0.7	+3.1	+7.0
Metals	Aluminum	3,373	+10.8	+31.3	+68.9

Energy: Scenario Fan Charts

Energy Commodities — Scenario Fan Charts (Channel A)

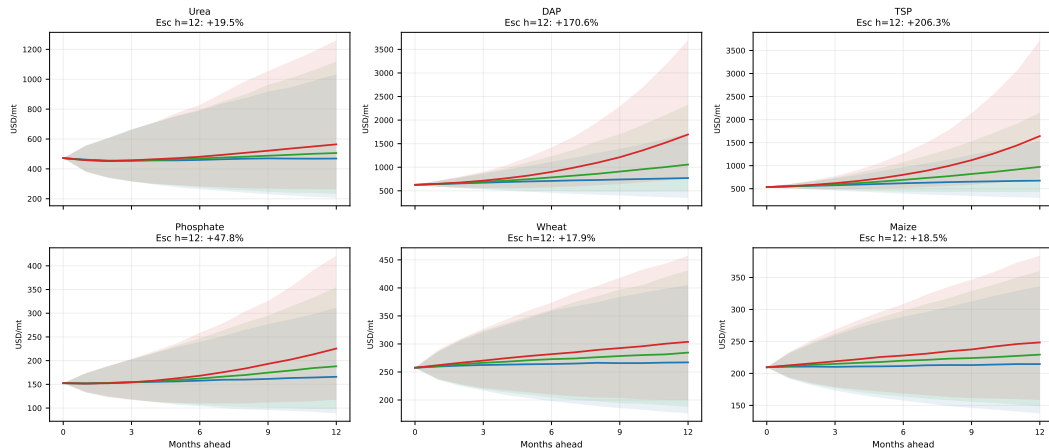
De-escalation Baseline Escalation



Fertilizers & Agriculture: Scenario Fan Charts

Fertilizers & Agriculture — Scenario Fan Charts (Channel A)

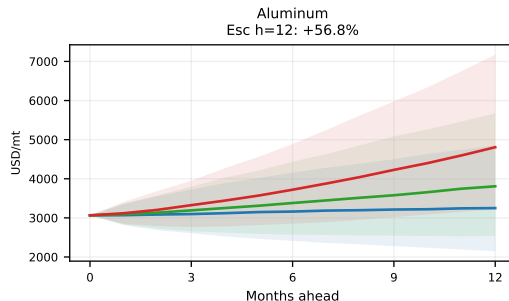
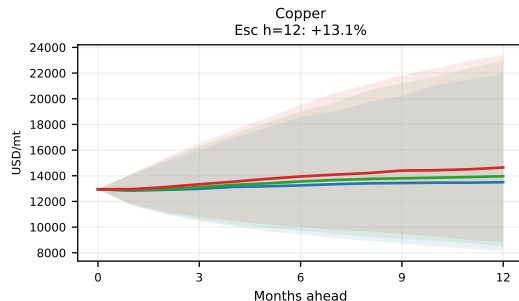
De-escalation Baseline Escalation



Metals: Scenario Fan Charts

Metals — Scenario Fan Charts (Channel A)

De-escalation Baseline Escalation



Scenario Fan Charts: Key Takeaways

Energy:

- Crude oil and Brent separate clearly across scenarios — direct GPR sensitivity ($\hat{\kappa}^T$ significant at 1%)
- LNG Japan: scenario paths converge — no significant GPR coefficient; price driven by long-term contracts
- Coal AU: +70.7% under escalation via AR persistence and energy substitution demand

Fertilizers & Agriculture:

- DAP and TSP diverge sharply (+256%, +305%) via mutual cross-beta amplification — not direct GPR sensitivity ($\hat{\kappa}^T \approx 0$)
- Wheat and Maize show real but moderate scenario spread

Metals:

- Aluminum: strongest direct GPR sensitivity ($p = 0.003$); +68.9% under escalation via energy-intensive electrolytic smelting
- Copper: marginal GPR sensitivity; moderate escalation response

Fertilizer Indirect Transmission: Network Decomposition

Escalation vs Baseline, $h = 12$ — pp gap (Esc – Base) decomposed into GPR-direct vs network-propagated components (v5_koyck_mar2026; network/direct split uses v4 structural shares):

Commodity	Full (pp)	Direct GPR (pp)	Network (pp)	Net share
CrudeOil ***	+19.9	+16.9	+3.0	15%
Brent ***	+23.3	+20.3	+3.0	13%
Aluminum **	+37.6	+26.0	+11.7	31%
Coal AU	+35.4	+34.0	+1.4	4%
NatGas EU	+30.8	+24.3	+6.5	21%
DAP	+139.9	+36.4	+103.5	74%
TSP	+172.8	+50.1	+122.7	71%
Phosphate	+36.0	+2.2	+33.8	94%
Urea	+36.9	+14.4	+22.5	61%
Wheat	+10.2	+10.2	≈ 0	0%

Second-order energy shock

For DAP and TSP the network share is **74%** and **71%**; Phosphate reaches **94%**. Energy commodities show only 13–21% network share (mostly direct GPR). The fertilizer price spike is a delayed, amplified consequence of the energy shock — statistically grounded, not a model artefact.

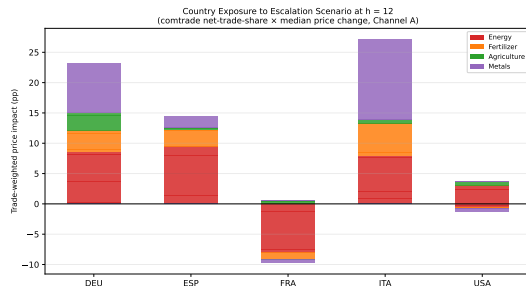
Country Exposure: Methodology

- **Source:** UN Comtrade bilateral trade flows, 2021–2023 three-year average
- **Net exposure:** (imports – exports) / total bilateral trade per commodity–country pair
- Positive net exposure = net importer: price rise is a cost to the trade balance
- Negative net exposure = net exporter: price rise is a gain
- Natural gas: consolidated into a single composite series (average of NatGas US and NatGas EU price paths) to avoid double-counting from HS code 2711 shared by both benchmarks
- **Exposure index:** $\sum_i w_i \cdot \hat{P}_i^{\text{Escalation}}$, where w_i is the net trade share

Scope

Trade-balance exposure captures the first-order macro impact under the escalation scenario — the commodity cost shock to each economy's import bill. Credit-risk and financial contagion channels are not modelled.

Country Exposure Under Escalation



Country	Role	Dominant exposure
DEU	Net energy importer	Large: NatGas EU, Coal, Fertilizers
ITA	Net energy importer	Similar to DEU; fertilizer-intensive
ESP	Net energy importer	Oil and gas dominant
FRA	Mixed	Oil importer, partially nuclear-insulated
USA	Net energy exporter	Negative energy bars offset import costs

Under escalation at $h = 12$, net-importing European economies face exposure-weighted commodity basket increases of +31.0% (DEU), +37.6% (ITA), and +36.4% (ESP). France, partially insulated by nuclear energy, registers a -23.8% net cost change: France is a large net agricultural exporter (wheat, maize) in the Comtrade basket, whose grain-export gains under rising prices more than offset its crude oil and gas import costs. The USA, a net energy exporter, faces only +6.2%. (v5_koyck_mar2026, Comtrade weights.)

Key Findings and Limitations

- Crude oil responds asymmetrically: GPRT threat raises prices (risk premium); GPRA act lowers them (demand destruction) — *buy the rumour, sell the news*
- **Fertilizers** are the most vulnerable commodity under escalation (DAP +256%, TSP +305%), driven predominantly by indirect network propagation (74% and 71% network shares), not direct GPR sensitivity
- Geopolitical crises **decouple** cross-sector correlation ($\hat{\beta}_T < 0$): the independence assumption is approximately correct for cross-sector pairs under stress
- Under escalation, European net importers face 30–40% commodity cost increases over 12 months (DEU +31%, ITA +38%, ESP +36%)

Limitations:

- Fertilizer channel relies on cross-beta network stability
- DCC ARCH ≈ 0 : correlation dynamics are regime-level, not shock-level
- GPR scenario paths are deterministic (AR(1) mean); GPR uncertainty is not propagated

The three-layer model is not a black-box forecast. Each layer produces an independent, interpretable finding about commodity price dynamics under geopolitical stress.

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